Temperature

If a change such as temperature is made to a reaction in equilibrium, the speed of the forward or backward reaction will adjust to counter the effects of that change. Every reversible reaction has an exothermic direction (giving out energy) and an endothermic one (taking in energy). If heated, the reaction that takes in heat (the endothermic direction) will speed up to balance the effect of the heating.

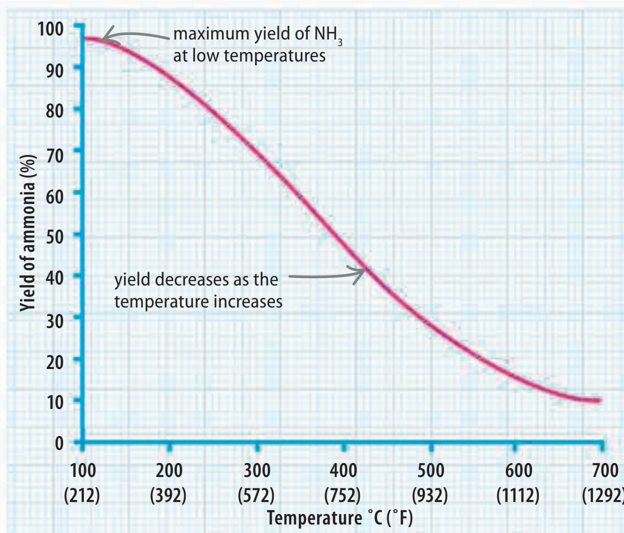
nitrogen gas

hydrogen gas

ammonia gas

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

▷ **Making ammonia**
Nitrogen and hydrogen give out heat when they react to form ammonia. Adding heat reduces the yield of ammonia because more of the compound decomposes back into hydrogen and nitrogen.



Pressure

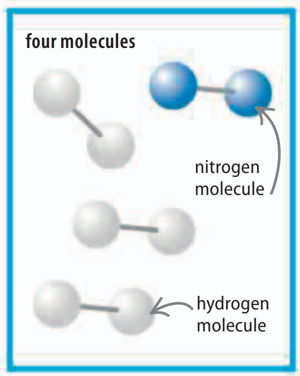
Pressure also affects the equilibrium in reactions involving gases. Pressure is caused by gas molecules hitting the sides of the container. The more molecules there are, the higher the pressure in the container. Increasing the pressure during a reversible reaction shifts the equilibrium toward the side with fewer molecules. In the equation for making ammonia there are four molecules of reactants (one molecule of nitrogen and three molecules of hydrogen) and two molecules of product (ammonia). An increase in pressure favors the forward direction of the reaction, which produces ammonia, rather than the reverse.

one molecule of nitrogen

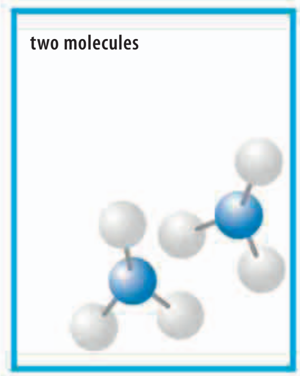
three molecules of hydrogen

two molecules of ammonia

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$



△ **Reactants**
Increasing the pressure pushes the hydrogen and nitrogen molecules together and drives the reaction to produce ammonia.



△ **Products**
Only two ammonia molecules are produced, which take up less space than the reactants. This makes the pressure fall.

Photosynthesis is a reversible reaction. If there is too much sugar and oxygen in a plant cell, **photorespiration** occurs, turning them back into carbon dioxide and water.

REAL WORLD

Quicklime

Quicklime is a chemical made by heating calcium carbonate. The heat makes the carbonate decompose into quicklime and carbon dioxide. However, these two products can then react back into calcium carbonate. To stop this, the kiln draws carbon dioxide away from the quicklime.